Data Format Description

This document explains the structure of the Multi-Depot Multi-Period Maintenance Routing Problem (MDMPMRP) instance files.

Each instance file is divided into four main sections:

[GENERAL_DATA]
[COST_MATRIX]
[TIME_MATRIX]
[ACTIVITIES]

The section headers are enclosed in square brackets.

Indexing Convention: All assets, depots, and activities are indexed starting from 1.

==
1. [GENERAL_DATA]
==

This section contains general parameters of the problem instance. Each line follows the format:

ParameterName: Value

The parameters are:

V
    Number of vehicles/teams.

M
    A large penalty coefficient used to discourage infeasible solutions during heuristic search.

Activities NO
    Number of maintenance activities/jobs in the instance.

n
    Total number of assets/locations including the depots. The number of actual assets excluding the depots is found by subtracting the number of depots from n. The title of the intance file writes this actual number of assets.

Primary Depot Index
    Index of the primary depot.

DepotsNO
    Number of depots in the instance.

T
    Length of the planning horizon, usually expressed in days.

WHours

Regular working hours per day.

Max_WHhours
Maximum allowable working hours per day, including overtime if applicable.

Co
Cost coefficient related to overtime or additional working time.

Cv
Fixed cost of using a vehicle/team.

Pu
Daily unavailability cost for the broken assets. An asset is unavailable for total difference of days between the start and end of the maintenance.

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2. [COST_MATRIX]
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This section gives the travelling cost between assets/locations.

The first row after the section header contains the asset indices. The last DepotsNO columns are the indices corresponding to the depots. Each subsequent row contains one row of the matrix. The last DepotsNO rows correspond to the depots.

Example structure:

```
[COST_MATRIX]
1 2 3 4
0 4 2 6
4 0 5 3
2 5 0 1
6 3 1 0
```

The entry in row i and column j represents the travelling cost from asset/location i to asset/location j.

Properties:
- The matrix is square.
- The size of the matrix depends on the instance.
- Diagonal values are typically zero.
- The matrices may be symmetric or asymmetric depending on the instance.

==
3. [TIME_MATRIX]
==

This section gives the travelling time (in hours) between assets/locations.

The format is the same as the cost matrix.

Example structure:

```
[TIME_MATRIX]
1 2 3 4
0 2 1 3
2 0 2 1
1 2 0 2
3 1 2 0
```

The entry in row i and column j represents the travelling time from asset/location i to asset/location j.

Properties:
- The matrix is square.
- The size of the matrix depends on the instance.
- Diagonal values are typically zero.
- The matrices may be symmetric or asymmetric depending on the instance.

==
4. [ACTIVITIES]
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This section contains activity-level data. The first row gives the column names. Each subsequent row corresponds to one activity or location entry. The last DepotsNO rows correspond to the depots.

The columns are:

Activity
    Activity identifier.

Asset
    Asset/location at which the activity is performed.

Maintenance Time (hr)
    Processing or maintenance duration of the activity in hours.

Release Date (Days)
    Earliest time (day) at which the activity can start.

Due Date (Days)
    Due date of the activity, expressed in days.

Penalty - Late
    Penalty coefficient for completing the activity after its due date.

Penalty - Early
    Penalty coefficient for completing the activity before its due date.

Broken
    Group identifier for broken activities. A value of 0 indicates that the activity does not belong to a broken group. Activities sharing the

same positive identifier belong to the same broken group and must be performed in increasing activity-index order (precedence).

TE

Binary indicator equal to 1 if the activity is the last activity in its broken group (or if the activity does not belong to any broken group), and 0 otherwise.

Example structure:

[ACTIVITIES]

Activity	Asset	Maintenance Time (hr)	Release Date	Due Date (Days)	Penalty - Late	Penalty - Early	Broken	TE
1	1	4	0	7	10	5	0	1
2	2	2	0	4	10	0	1	0
3	2	2	1	4	15	10	1	1
4	3	3	0	3	10	5	2	0
5	3	1	2	7	20	0	2	1
6	4	2	0	5	10	5	0	1

In this sample data, there are a total of 6 activities performed on 4 assets. Activity 2 and 3 belong to the same broken group, where 2 is a predecessor of 3. Since activity 3 is the last one of this broken group, TE value of activity 2 is 0, and activity 3 is 1. Similarly, Activities 4 and 5 belong to the same (different than the first one) broken group.

Notes:

- The number of rows changes depending on the instance.

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5. File Naming Convention

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Each file is named as:

aA_nN_DepP_dD_vV_BB_repR.txt

Example:

a12_n4_Dep3_d7_v2_B0_rep1.txt

The components indicate:

aA : number of maintenance activities
nN : number of assets that the activities belong to
DepP : number of depots
dD : planning horizon
vV : number of vehicles/teams

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BB     : number of broken acivities
repR   : random replication number
```

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6. Recommended Reading Procedure
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To read an instance file programmatically:

1. Read the file line by line.
2. Detect section headers:
   [GENERAL_DATA], [COST_MATRIX], [TIME_MATRIX], [ACTIVITIES].
3. Parse key-value pairs under [GENERAL_DATA].
4. Parse numerical matrices under [COST_MATRIX] and [TIME_MATRIX].
5. Parse tab-separated activity records under [ACTIVITIES].

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7. Reproducibility Notes
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- The text files are intended to support computational reproducibility.
- Users should verify that all model-specific parameter interpretations are consistent with the associated paper.